

Backend Networking Cheat Sheet (v2)

Big Picture

Client → HTTP Request → IP + Port → Operating System → Node.js/Express → Route → Business Logic → Database → HTTP Response → Client

IP Address

Identifies the computer. Think: hotel/house address.

Port

Identifies the application listening on the computer. Think: room number.

Route

Identifies the function inside the application (e.g. /movies, /predict).

Port vs Route

Port: Which application should receive the request? Route: Which function inside that application should handle it? Flow: Browser → localhost:8000 → OS → Express → /predict → Prediction code.

HTTP Request

Request line + Headers + optional Body.

Headers

Metadata/instructions (Content-Type, Authorization, Accept).

Body / Payload

Actual data (often JSON). Think: the letter inside the envelope.

GET vs POST

GET retrieves an existing resource (may include filters). POST submits data for processing or creates a resource. GET/POST are always described from the CLIENT perspective; the server sends an HTTP response.

Common Ports

80 HTTP, 443 HTTPS, 3000 React, 5432 PostgreSQL, 6379 Redis, 8000 Backend, 27017 MongoDB.

Analogies

IP=Hotel address. Port=Room number. Route=Office inside the building. Headers=Envelope instructions. Body=Letter.